

# LESSON 7

## TRUTH TABLES, EQUIVALENT STATEMENTS, AND TAUTOLOGIES

### Overview:

A logical statement is either true or false. This is how we distinguish a logical statement among others. Identifying the truth value of a given statement is the first part of the lesson. This lesson will provide students with the knowledge on how to determine the truth value of a statement whether it is a negation, conjunction, disjunction, conditional, or biconditional statements by means of constructing truth tables. Equivalent statements, tautologies and fallacies are also discussed in this section.

### Learning Outcomes:

At the end of Lesson 7, the students can:

1. identify the truth value of a statement using truth tables;
2. construct a truth table for given statement;
3. show that two statements are equivalent; and
4. prove that a given statement is a tautology or a fallacy.

### Materials Needed:

- slide presentation
- hand-outs on the different truth tables of statement
- video lesson

**Duration:** 3 hours

### Learning Content:

#### 7.1 TRUTH TABLES

**Truth tables** represent the relationships between the truth values of statements and compound statements drawn from the given statements.

**Note:** A Compound statement that involves  $n$  simple statements required a standard truth table form with  $2^n$  rows.

| $P$ | $q$ | Given statement |
|-----|-----|-----------------|
| T   | T   |                 |
| T   | F   |                 |
| F   | T   |                 |
| F   | F   |                 |

Standard truth table form for a statement involving two simple statements

$2^n = 2^2 = 4$  rows

| $p$ | $q$ | $r$ | Given Statement |
|-----|-----|-----|-----------------|
| T   | T   | T   |                 |
| T   | T   | F   |                 |
| T   | F   | T   |                 |
| T   | F   | F   |                 |
| F   | T   | T   |                 |
| F   | T   | F   |                 |
| F   | F   | T   |                 |
| F   | F   | F   |                 |

Standard truth table form for a statement involving three simple statements

$2^n = 2^3 = 8$  rows

Table 7.1

Truth Table for Negation

| $P$ | $\sim p$ |
|-----|----------|
| T   | F        |
| F   | T        |

**Truth Value of a NEGATION**  
 If the statement  $p$  is true, its negation  $\sim p$  is false; otherwise if  $p$  is false,  $\sim p$  is true.

EXAMPLE 1

Write the negation of the statement “I want to watch a movie.”.

Solution

The negation of the statement “I want to watch a movie.” is “I don’t want to watch a movie.”

Table 7.2

Truth Table for Conjunction

| $P$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

**Truth Value of a CONJUNCTION**  
 The conjunction  $p \wedge q$  is true if and only if both  $p$  and  $q$  are true.

**EXAMPLE 2**

Determine whether the statement “An even number ends with a 2 and it is divisible by 2.” Is true or false.

*Solution*

This is a true statement since it is both true that an even number ends with a 2 and an even number is divisible by 2.

**EXAMPLE 3**

Is the statement “1 is a prime number and 1 is an odd number.” true?

*Solution*

1 is neither a prime nor a composite number but 1 is an odd number. Hence the given is a false statement.

**Table 7.3**  
Truth Table for Disjunction

| <i>P</i> | <i>q</i> | <i>p</i> $\wedge$ <i>q</i> |
|----------|----------|----------------------------|
| T        | T        | T                          |
| T        | F        | F                          |
| F        | T        | F                          |
| F        | F        | F                          |

**Truth Value of a DISJUNCTION**  
The disjunction  $p \vee q$  is true if and only if  $p$  is true,  $q$  is true, or both  $p$  and  $q$  are true.

**EXAMPLE 4**

Determine whether the statement “2 is an odd number or 2 is prime.” is true or false.

*Solution*

The statement “2 is an odd number.” is definitely a false statement. However, it is true that “2 is prime”. Thus the given statement is true.

**EXAMPLE 5**

Is  $-5 \geq -2$  true?

*Solution*

$-5 \geq -2$  means  $-5 > -2$  or  $-5 = -2$ . One of these statements must be true for the original statement to be true. Because  $-5 \nless -2$  and  $-5 \neq -2$ , the original statement is false.

**Table 7.4**

Truth Table for Conditional

| <i>P</i> | <i>q</i> | $p \rightarrow q$ |
|----------|----------|-------------------|
| T        | T        | T                 |
| T        | F        | F                 |
| F        | T        | T                 |
| F        | F        | T                 |

**Truth Value of a CONDITIONAL**

The conditional  $p \rightarrow q$  is false if  $p$  is true and  $q$  is false. It is true in all other cases.

In the conditional  $p \rightarrow q$ ,  $p$  is the antecedent and  $q$  is the consequent.

**EXAMPLE 6**

Determine whether the statement “If 6 is an even number, then 6 is divisible by 2.” is true or false.

*Solution*

Because the consequent is true, thus the statement is true.

**Table 7.5**

Truth Table for Biconditional

| <i>P</i> | <i>q</i> | $p \leftrightarrow q$ |
|----------|----------|-----------------------|
| T        | T        | T                     |
| T        | F        | F                     |
| F        | T        | F                     |
| F        | F        | T                     |

**Truth Value of a BICONDITIONAL**

The biconditional  $p \leftrightarrow q$  is true if both  $p$  and  $q$  is true or both  $p$  and  $q$  is false. If either one is false, the biconditional is false.

**EXAMPLE 7**

Is the statement “3 is an odd number if and only if 3 is not divisible by 2.” true?

*Solution*

Both statements “3 is an odd number.” and “3 is not divisible by 2.” are true. Hence, the biconditional statement is true.

**EXAMPLE 8**

Construct a truth table for  $\sim(\sim p \vee q) \wedge q$ .

*Solution*

Start with the standard truth table form and then include  $\sim p$  column.

| <i>P</i> | <i>q</i> | $\sim p$ |
|----------|----------|----------|
| T        | T        | F        |
| T        | F        | F        |
| F        | T        | T        |
| F        | F        | T        |

Now use the truth values from  $\sim p$  and  $q$  columns to produce the truth values for  $\sim p \vee q$ , as shown in the rightmost column of the following table.

| $p$ | $q$ | $\sim p$ | $\sim p \vee q$ |
|-----|-----|----------|-----------------|
| T   | T   | F        | T               |
| T   | F   | F        | F               |
| F   | T   | T        | T               |
| F   | F   | T        | T               |

Negate the truth values of  $\sim p \vee q$  column to produce the following.

| $p$ | $q$ | $\sim p$ | $\sim p \vee q$ | $\sim (\sim p \vee q)$ |
|-----|-----|----------|-----------------|------------------------|
| T   | T   | F        | T               | F                      |
| T   | F   | F        | F               | T                      |
| F   | T   | T        | T               | F                      |
| F   | F   | T        | T               | F                      |

For the last step, find the truth values of the conjunction  $\sim (\sim p \vee q) \wedge q$  and place the results in the rightmost column of the table. The shaded column is the truth table of the compound statement  $\sim (\sim p \vee q) \wedge q$ .

| $p$ | $q$ | $\sim p$ | $\sim p \vee q$ | $\sim (\sim p \vee q)$ | $\sim (\sim p \vee q) \wedge q$ |       |
|-----|-----|----------|-----------------|------------------------|---------------------------------|-------|
| T   | T   | F        | T               | F                      | F                               | row 1 |
| T   | F   | F        | F               | T                      | F                               | row 2 |
| F   | T   | T        | T               | F                      | F                               | row 3 |
| F   | F   | T        | T               | F                      | F                               | row 4 |

### EXAMPLE 9

Use the truth table from example 8 to determine the truth value of  $\sim (\sim p \vee q) \wedge q$  given that  $p$  is true,  $q$  is false.

*Solution*

In row 2 of the previous table, we see that when  $p$  is true and  $q$  is false, the statement  $\sim (\sim p \vee q) \wedge q$  in the rightmost column is false.

### EXAMPLE 10

Construct a truth table for  $(p \rightarrow \sim r) \wedge (q \leftrightarrow p)$ .

*Solution*

Using the procedures developed in example 8, we can produce the following table. Notice that we have 8 rows since we have three simple statements  $p$ ,  $q$ , and  $r$ .

| $p$ | $q$ | $R$ | $\sim r$ | $p \rightarrow \sim r$ | $q \leftrightarrow p$ | $(p \rightarrow \sim r) \wedge (q \leftrightarrow p)$ |       |
|-----|-----|-----|----------|------------------------|-----------------------|-------------------------------------------------------|-------|
| T   | T   | T   | F        | F                      | T                     | F                                                     | row 1 |
| T   | T   | F   | T        | T                      | T                     | T                                                     | row 2 |
| T   | F   | T   | F        | F                      | F                     | F                                                     | row 3 |
| T   | F   | F   | T        | T                      | F                     | F                                                     | row 4 |
| F   | T   | T   | F        | T                      | F                     | F                                                     | row 5 |
| F   | T   | F   | T        | T                      | F                     | F                                                     | row 6 |
| F   | F   | T   | F        | T                      | T                     | T                                                     | row 7 |
| F   | F   | F   | T        | T                      | T                     | T                                                     | row 8 |

1

2

3

4

7.2 EQUIVALENT STATEMENTS

Two statements are said to be **logically equivalent** if they both have exactly the same truth values for all possible truth values of their simple statements. This implies that the two statements have exactly the same meaning. For EXAMPLE, the statement “Today is December 25 and today is holiday.” is logically equivalent to “Today is holiday and today is December 25.”. Similarly, “I have a dog or I have a cat.” is the same by saying “I have a cat or I have a dog.”.

Logical equivalence is denoted by the symbol:  $\equiv$ .

EXAMPLE 11

Show that  $\sim (p \vee q)$  and  $\sim p \wedge \sim q$  are equivalent.

Solution

We construct the truth tables of each statement as shown below and compare the rightmost columns of the two statements.

Truth table for  $\sim (p \vee q)$  :

| $P$ | $q$ | $p \vee q$ | $\sim (p \vee q)$ |
|-----|-----|------------|-------------------|
| T   | T   | T          | F                 |
| T   | F   | T          | F                 |
| F   | T   | T          | F                 |
| F   | F   | F          | T                 |

12

Truth table for  $\sim p \wedge \sim q$ :

| $p$ | $q$ | $\sim p$ | $\sim q$ | $\sim p \wedge \sim q$ |
|-----|-----|----------|----------|------------------------|
| T   | T   | F        | F        | F                      |
| T   | F   | F        | T        | F                      |
| F   | T   | T        | F        | F                      |
| F   | F   | T        | T        | T                      |

123

Identical truth values

Notice that the last columns on both statements, the shaded ones, have the same truth values F,F,F, and T. Thus,  $\sim (p \vee q)$  and  $\sim p \wedge \sim q$  are equivalent. In symbols,  $\sim (p \vee q) \equiv \sim p \wedge \sim q$ .

**EXAMPLE 12**

Show that  $\sim (p \wedge q)$  and  $\sim p \vee \sim q$  are equivalent.

*Solution*

We construct the truth tables of each statement as shown below and compare the rightmost columns of the two statements.

Truth table for  $\sim (p \wedge q)$  :

| $P$ | $q$ | $p \wedge q$ | $\sim (p \wedge q)$ |
|-----|-----|--------------|---------------------|
| T   | T   | T            | F                   |
| T   | F   | F            | T                   |
| F   | T   | F            | T                   |
| F   | F   | F            | T                   |

1

2



Truth table for  $\sim p \vee \sim q$ :

| $p$ | $q$ | $\sim p$ | $\sim q$ | $\sim p \vee \sim q$ |
|-----|-----|----------|----------|----------------------|
| T   | T   | F        | F        | F                    |
| T   | F   | F        | T        | T                    |
| F   | T   | T        | F        | T                    |
| F   | F   | T        | T        | T                    |

1

2

3

Identical truth values



Notice that the last columns on both statements have the same truth values. Thus,  $\sim (p \wedge q)$  and  $\sim p \vee \sim q$  are equivalent. In symbols,  $\sim (p \wedge q) \equiv \sim p \vee \sim q$ .

The two previous examples of equivalence are known as **De Morgan’s Laws** for statements.

**De Morgan’s Laws for Statements**

For any statements  $p$  and  $q$ ,

$$\begin{aligned} \sim (p \vee q) &\equiv \sim p \wedge \sim q \\ \sim (p \wedge q) &\equiv \sim p \vee \sim q \end{aligned}$$

These laws can be used to restate an English sentence in an equivalent form.

**EXAMPLE 13**

Restate the following statement using one of the De Morgan’s Laws for statements.

“It is not true that, I went to the market and I bought some fruits.”

*Solution*

Let the  $p$  represent the statement “I went to the market.”. Let  $q$  represent the statement “I bought some fruits.”. The given statement in symbolic form is  $\sim (p \wedge q)$  and we know that  $\sim (p \wedge q)$  is logically equivalent to  $\sim p \vee \sim q$ . Hence, the

original sentence is equivalent to “I did not go to the market or I did not buy some fruits.”

### 7.3 TAUTOLOGIES AND SELF – CONTRADICTIONS

A **tautology** is a statement that is always true. A **self-contradiction** is a statement that is always false.

**EXAMPLE 14**

Show that  $d \vee (\sim d \vee e)$  is a tautology.

*Solution*

| $d$ | $e$ | $\sim d$ | $\sim d \vee e$ | $d \vee (\sim d \vee e)$ |
|-----|-----|----------|-----------------|--------------------------|
| T   | T   | F        | T               | T                        |
| T   | F   | F        | F               | T                        |
| F   | T   | T        | T               | T                        |
| F   | F   | T        | T               | T                        |

1

2

3

The shaded column shows that  $d \vee (\sim d \vee e)$  is always true. Thus  $d \vee (\sim d \vee e)$  is a tautology.

**EXAMPLE 15**

Show that  $d \wedge (\sim d \wedge e)$  is a fallacy.

*Solution*

| $d$ | $e$ | $\sim d$ | $\sim d \wedge e$ | $d \wedge (\sim d \wedge e)$ |
|-----|-----|----------|-------------------|------------------------------|
| T   | T   | F        | F                 | F                            |
| T   | F   | F        | F                 | F                            |
| F   | T   | T        | T                 | F                            |
| F   | F   | T        | F                 | F                            |

1

2

3

The shaded column shows that  $d \wedge (\sim d \wedge e)$  is always false. Thus  $d \wedge (\sim d \wedge e)$  is a self-contradiction.



## Learning Activity

**Direction:** Let the students work by pair. Each pair will be given two compound statements. Each student of each pair will pick a statement and construct its truth table. The two students will then compare the truth tables they have constructed and state the characteristics of the statements they have chosen (whether the two statements are equivalent, one is a tautology or fallacy, or both of them are tautologies or self-contradictions).

- |                                                             |                                                                     |
|-------------------------------------------------------------|---------------------------------------------------------------------|
| 1. $p \vee q$ and $q \vee p$                                | 6. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$       |
| 2. $p \wedge q$ and $q \wedge p$                            | 7. $\sim [(p \vee q) \vee r]$ and $\sim (p \vee q) \wedge \sim r$   |
| 3. $p \vee (q \vee r)$ and $(p \vee q) \vee r$              | 8. $\sim [(p \vee q) \wedge r]$ and $\sim (p \vee q) \vee \sim r$   |
| 4. $p \wedge (q \wedge r)$ and $(p \wedge q) \wedge r$      | 9. $\sim (p \vee \sim p) \wedge q$ and $p \wedge (\sim p \wedge q)$ |
| 5. $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ | 10. $\sim (p \wedge \sim p) \vee q$ and $p \vee (\sim p \vee q)$    |

## Learning Evaluation

A. For each of the following statements, construct a truth table and state whether the statement is a tautology, self-contradiction, or neither.

- |                                                                    |                                                             |
|--------------------------------------------------------------------|-------------------------------------------------------------|
| 1. $p \wedge \sim p$                                               | 6. $(p \rightarrow q) \leftrightarrow (q \rightarrow p)$    |
| 2. $p \vee \sim p$                                                 | 7. $p \rightarrow (\sim q \vee r)$                          |
| 3. $(p \wedge \sim q) \rightarrow q$                               | 8. $\sim p \rightarrow (q \rightarrow r)$                   |
| 4. $(p \vee q) \rightarrow (p \wedge q)$                           | 9. $(p \rightarrow q) \vee (\sim p \rightarrow r)$          |
| 5. $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ | 10. $(p \leftrightarrow q) \vee (\sim q \leftrightarrow r)$ |

B. Use truth tables to determine which of the following are equivalent to each other:

- |                                               |                                             |
|-----------------------------------------------|---------------------------------------------|
| 1. $p$                                        | 5. $(p \vee \sim q) \wedge (q \vee \sim p)$ |
| 2. $\sim (\sim p)$                            | 6. $\sim (p \vee q)$                        |
| 3. $(p \wedge q) \vee (\sim p \wedge \sim q)$ | 7. $p \rightarrow q$                        |
| 4. $\sim p \vee q$                            | 8. $(q \wedge p) \vee \sim p$               |

## References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.